POWER SUPPLY

POWER SUPPLY

File: POWER SUPPLY.kicad_sch

USB CONNECTOR

USB2.0

USB_ID
TYPE_C_D+
TYPE_C_D−

File: USB CONNECTOR.kicad_sch

GPIO

GPIO

USB2_D+
USB2_D−

File: KEYPAD.kicad_sch

AUDIO IO

AUDIO COdeC

USB1_D+
USB1_D−

File: AUDIO IO.kicad_sch

USB HUB

USB HUB

USB_ID
TYPE_C_D+
TYPE_C_D−
USB2_D+
USB2_D−
USB1_D+
USB1_D−

File: USB HUB.kicad_sch

CM5

COMPUTE MODULE

CM5_OTG_ID
USB_CM5_D+
USB_CM5_D−
nEXTRST
SCL
SDA
TOUCH_INT
DISP_CSD
PCLK_SDA
DE_SCL
VSYNC
HSYNC
R2
R3
R4
R5
R6
R7
G2
G3
G4
G5
G6
G7
B2
B3
B4
B5
B6
B7
DISPLAY_BACKLIT
PCIE_TX_PD
PCIE_TX_ND
PCIE_RX_P
PCIE_RX_N
PCIE_CLK_PD
PCIE_CLK_ND
PCIE_nRSTD
PCIE_nCLKREQ
PCIE_nWAKE
PCIE_PWR_END
VBUS_END
USB3-0-D_P
USB3-0-D_N
USB3-0-TX_P
USB3-0-TX_N
USB3-0-RX_P
USB3-0-RX_N
USB3-1-D_P
USB3-1-D_N
USB3-1-TX_P
USB3-1-TX_N
USB3-1-RX_P
USB3-1-RX_N

File: CM4.kicad_sch

DISPLAY

DISPLAY

SCL
SDA
TOUCH_INT
DISP_CS
PCLK_SDA
DE_SCL
VSYNC
HSYNC
R2
R3
R4
R5
R6
R7
G2
G3
G4
G5
G6
G7
B2
B3
B4
B5
B6
B7
DISPLAY_BACKLIT

File: DISPLAY.kicad_sch

USB & PCIe

PCIe

PCIE_TX_P
PCIE_TX_N
PCIE_RX_P
PCIE_RX_N
PCIE_CLK_P
PCIE_CLK_N
PCIE_nRST
PCIE_nCLKREQ
PCIE_nWAKE
PCIE_PWR_EN
VBUS_EN
USB3-0-D_P
USB3-0-D_N
USB3-0-TX_P
USB3-0-TX_N
USB3-0-RX_P
USB3-0-RX_N
USB3-1-D_P
USB3-1-D_N
USB3-1-TX_P
USB3-1-TX_N
USB3-1-RX_P
USB3-1-RX_N

File: USB-PCIe.kicad_sch

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Sheet: /
File: PunkT_PCB.kicad_sch

Title: Cyber.T

Size: A3	Date: 2025-08-22	**Rev: 3.0**
KiCad E.D.A. 9.0.5		Id: 1/9

U6A
RPi-CM5

2	GND	GND	1
4	Ethernet_Pair1_P	Ethernet_Pair3_P	3
6	Ethernet_Pair1_N	Ethernet_Pair3_N	5
8	GND	GND	7
10	Ethernet_Pair0_N	Ethernet_Pair2_N	9
12	Ethernet_Pair0_P	Ethernet_Pair2_P	11
14	GND	GND	13
16	FAN_TACHO	Ethernet_nLED3(3.3v)	15
18	Ethernet_SYNC_OUT(3.3v)	Ethernet_nLED2(3.3v)	17
20	EEPROM_nWP	FAN_PWM	19
22	GND	LED_nACT	21 — ACT_LED
24	GPIO26	GND	23
26 — DISPLAY_BACKLIT	GPIO19	GPIO21	25 — B3
28 — G3	GPIO13	GPIO20	27 — B2
30 — R4	GPIO6	GPIO16	29 — G6
32	GND	GPIO12	31 — G2
34 — R3	GPIO5	GND	33
36 — PCLK_SDA	ID_SD	ID_SC	35 — DE_SCL
38 — SCLD	GPIO11	GPIO7	37 — R5
40 — R7	GPIO9	GPIO8	39 — R6
42	GND	GPIO25	41 — B7
44 — SDAD	GPIO10	GND	43
46 — B4	GPIO22	GPIO24	45 — B6
48 — TOUCH_INT	GPIO27	GPIO23	47 — B5
50 — G7	GPIO17	GPIO18	49 — DISP_CS
52	GND	GPIO15	51 — G5
54 — R2	GPIO4	GND	53
56 — HSYNCD	GPIO3	GPIO14	55 — G4
58 — VSYNCD	GPIO2	SD_CLK	57 — SD_CLK
60	GND	GND	59
62 — SD_CMD	SD_CMD	SD_DAT3	61 — SD_D3
64	SD_DAT5	SD_DAT0	63 — SD_D0
66	GND	GND	65
68	SD_DAT4	SD_DAT1	67 — SD_D1
70	SD_DAT7	SD_DAT2	69 — SD_D2
72	SD_DAT6	GND	71
74	GND	SD_VDD_Override	73
76	VBAT	SD_PWR_ON	75
78	GPIO_VREF(1.8v/3.3v_Input)	+5v_(Input)	77 — +5V
80 — SCL0	SCL0	+5v_(Input)	79
82 — SDA0	SDA0	+5v_(Input)	81
84	+3.3v_(Output)600mA Max	+5v_(Input)	83
86	+3.3v_(Output)	+5v_(Input)	85
88	+1.8v_(Output)600mA Max	+5v_(Input)	87
90	+1.8v_(Output)	WiFi_nDisable	89
92 — PWR_BUT	PWR_BUT	BT_nDisable	91
94	CC1	nRPIBOOT	93 — TP12
96	CC2	LED_nPWR	95 — nPWR_LED
98	GND	CAM_GPIO0	97 — CAM_GPIO0
100 — CAM_GPIO1	CAM_GPIO1	PMIC_ENABLE	99

GPIO

NB SD signals are only available on modules without eMMC

+3.3V — R42 0R

GND GND

When grounded the CM5 becomes a USB host

LOW >> HOST
HIGH >> deVICE

U6B
RPi-CM5

102 — PCIE_nCLKREQD	PCIe_nCLKREQ	USB_OTG_ID	101 — CM5_OTG_ID
104 — PCIE_nWAKED	PCIE_nWAKE	USB2_N	103 — USB_CM5_D-
106 — PCIE_PWR_EN	PCIE_PWR_EN	USB2_P	105 — USB_CM5_D+
108	GND	GND	107
110 — PCIE_CLK_P	PCIe_CLK_P	PCIe_nRST	109 — PCIE_nRST
112 — PCIE_CLK_N	PCIe_CLK_N	VBUS_EN	111 — VBUS_EN
114	GND	GND	113
116 — PCIE_RX_PD	PCIe_RX_P	MIPI0_D0_N	115 — CAM1_D0_N
118 — PCIE_RX_ND	PCIe_RX_N	MIPI0_D0_P	117 — CAM1_D0_P
120	GND	GND	119
122 — PCIE_TX_P	PCIe_TX_P	MIPI0_D1_N	121 — CAM1_D1_N
124 — PCIE_TX_N	PCIe_TX_N	MIPI0_D1_P	123 — CAM1_D1_P
126	GND	GND	125
128 — USB3-0-RX_ND	USB3-0-RX_N	MIPI0_C_N	127 — CAM1_C_N
130 — USB3-0-RX_PD	USB3-0-RX_P	MIPI0_C_P	129 — CAM1_C_P
132	GND	GND	131
134 — USB3-0-D_PD	USB3-0-D_P	MIPI0_D2_N	133 — CAM1_D2_N
136 — USB3-0-D_ND	USB3-0-D_N	MIPI0_D2_P	135 — CAM1_D2_P
138	GND	GND	137
140 — USB3-0-TX_ND	USB3-0-TX_N	MIPI0_D3_N	139 — CAM1_D3_N
142 — USB3-0-TX_PD	USB3-0-TX_P	MIPI0_D3_P	141 — CAM1_D3_P
144	GND	HDMI1_HOTPLUG	143
146	HDMI1_TX2_P	HDMI1_SDA	145
148	HDMI1_TX2_N	HDMI1_SCL	147
150	GND	HDMI1_CEC	149
152	HDMI1_TX1_P	HDMI0_CEC	151 — HDMI0_CEC
154	HDMI1_TX1_N	HDMI0_HOTPLUG	153 — HDMI0_HP
156	GND	GND	155
158	HDMI1_TX0_P	USB3-1-RX_N	157 — USB3-1-RX_N
160	HDMI1_TX0_N	USB3-1-RX_P	159 — USB3-1-RX_P
162	GND	GND	161
164	HDMI1_CLK_P	USB3-1-D_P	163 — USB3-1-D_P
166	HDMI1_CLK_N	USB3-1-D_N	165 — USB3-1-D_N
168	GND	GND	167
170 — HDMI0_TX2_P	HDMI0_TX2_P	USB3-1-TX_N	169 — USB3-1-TX_N
172 — HDMI0_TX2_N	HDMI0_TX2_N	USB3-1-TX_P	171 — USB3-1-TX_P
174	GND	GND	173
176 — HDMI0_TX1_P	HDMI0_TX1_P	MIPI1_D0_N	175 — DSI1_D0_N
178 — HDMI0_TX1_N	HDMI0_TX1_N	MIPI1_D0_P	177 — DSI1_D0_P
180	GND	GND	179
182 — HDMI0_TX0_P	HDMI0_TX0_P	MIPI1_D1_N	181 — DSI1_D1_N
184 — HDMI0_TX0_N	HDMI0_TX0_N	MIPI1_D1_P	183 — DSI1_D1_P
186	GND	GND	185
188 — HDMI0_CLK_P	HDMI0_CLK_P	MIPI1_C_N	187 — DSI1_CLK_N
190 — HDMI0_CLK_N	HDMI0_CLK_N	MIPI1_C_P	189 — DSI1_CLK_P
192	GND	GND	191
194 — DSI1_D3_N	MIPI1_D3_N	MIPI1_D2_N	193 — DSI1_D2_N
196 — DSI1_D3_P	MIPI1_D3_P	MIPI1_D2_P	195 — DSI1_D2_P
198	GND	GND	197
200 — HDMI0_SCL	HDMI0_SCL	HDMI0_SDA	199 — HDMI0_SDA

High Speed Serial

GND GND

+5V — F1 200mA — +5V

J2
HDMI_A_1.4
HC-HDMI0019T-4

HDMI0_TX2_P	1	D2+
HDMI0_TX2_N	3	D2-
HDMI0_TX1_P	4	D1+
HDMI0_TX1_N	6	D1-
HDMI0_TX0_P	7	D0+
HDMI0_TX0_N	9	D0-
HDMI0_CLK_P	10	CK+
HDMI0_CLK_N	12	CK-
HDMI0_CEC	13	CEC
HDMI0_SCL	15	SCL
HDMI0_SDA	16	SDA
HDMI0_HP	19	UTILITY/HEAC+ HPD/HEAC-
	18	+5V
	2, 5, 8, 11, 17, SH	D2S, D1S, D0S, CKS, GND, SH

GND

HDMI connector

SW4

DE_SCL 1 — 1 — 4 — 4 PCLK_SDA

+3.3V — R34 2.2k — 2 — 2 — 5 — 5 — R35 2.2k — +3.3V

3 — 3 — 6 — 6

+3.3V — C2 10uF — GND

J5
Micro_SD_Card
TF-01A

SD_D2	1	DAT2
SD_D3	2	DAT3/CD
SD_CMD	3	CMD
	4	VDD
SD_CLK	5	CLK
	6	VSS
SD_D0	7	DAT0
SD_D1	8	DAT1
	9	SHIELD

J15 CM5 — J16 CM5 — J20 display — GND — GND

MICRO SD CARD

J14
DSI 1
MIPI-DSI
X05A20L22T

DSI1_D0_N	1	2
DSI1_D0_P	3	4
DSI1_D1_N	5	6
DSI1_D1_P	7	8
DSI1_CLK_N	9	10
DSI1_D2_N	11	12
DSI1_D2_P	13	14
DSI1_D3_N	15	16
DSI1_D3_P	17	18
DE_SCL	19	20
PCLK_SDA	21	22

+3.3V — R21 2.2k — +3.3V — +3.3V — MP — GND GND

MIPI DSI
100R Differentail
50R Single Ended
inter and intra pair match 5mil

Impedance stackup JLC3313

J3
CSI 1
CAMERA
X05A20L22T

CAM1_D0_N	1	2
CAM1_D0_P	3	4
CAM1_D1_N	5	6
CAM1_D1_P	7	8
CAM1_C_N	9	10
CAM1_C_P	11	12
CAM1_D2_N	13	14
CAM1_D2_P	15	16
CAM1_D3_N	17	18
CAM1_D3_P	19	20
CAM_GPIO0	21	22
CAM_GPIO1		

+3.3V — SCL0 — SDA0 — MP — GND GND

MIPI CSI
100R Differentail
50R Single Ended
inter and intra pair match 5mil

Impedance stackup JLC3313

+3.3V — D3 Pi ACT — R3 1k — ACT_LED

+3.3V — D4 RPI PWR — R4 1k — nPWR_LED

Rpi Power status LED and Activity LED

PWR_BUT — SW1 PI WAKEUP — GND

RPi Wakeup switch

TP11 — nEXTRST

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Sheet: /CM5/		
File: CM4.kicad_sch		
Title: Cyber.T		
Size: A3	Date: 2025-08-22	**Rev: 3.0**
KiCad E.D.A. 9.0.5		Id: 2/9

When grounded the CM5 becomes a USB host

LOW >> HOST
HIGH >> deVICE

ID

LOW = FUSB303B attached as a Source (HOST)
HIGH-Z = FUSB303B attached as a Sink (Device)

USB port automatic role selection switch

Pos.1 – Allows USB port controller to select the role (Full automatic)
Pos.2 – Manually selects the DEVICE Mode
Pos.3 – Manually selects the HOST Mode

USB switches for switching USB connections

USB Switch for CM5

USB Switch for USB HUB

S = LOW >> D = D1
S = HIGH >> D = D2

ID

LOW = FUSB303B attached as a Source (HOST)
HIGH-Z = FUSB303B attached as a Sink (Device)

Auxiliary JST header for access to the USB hub when RPi is in device mode.

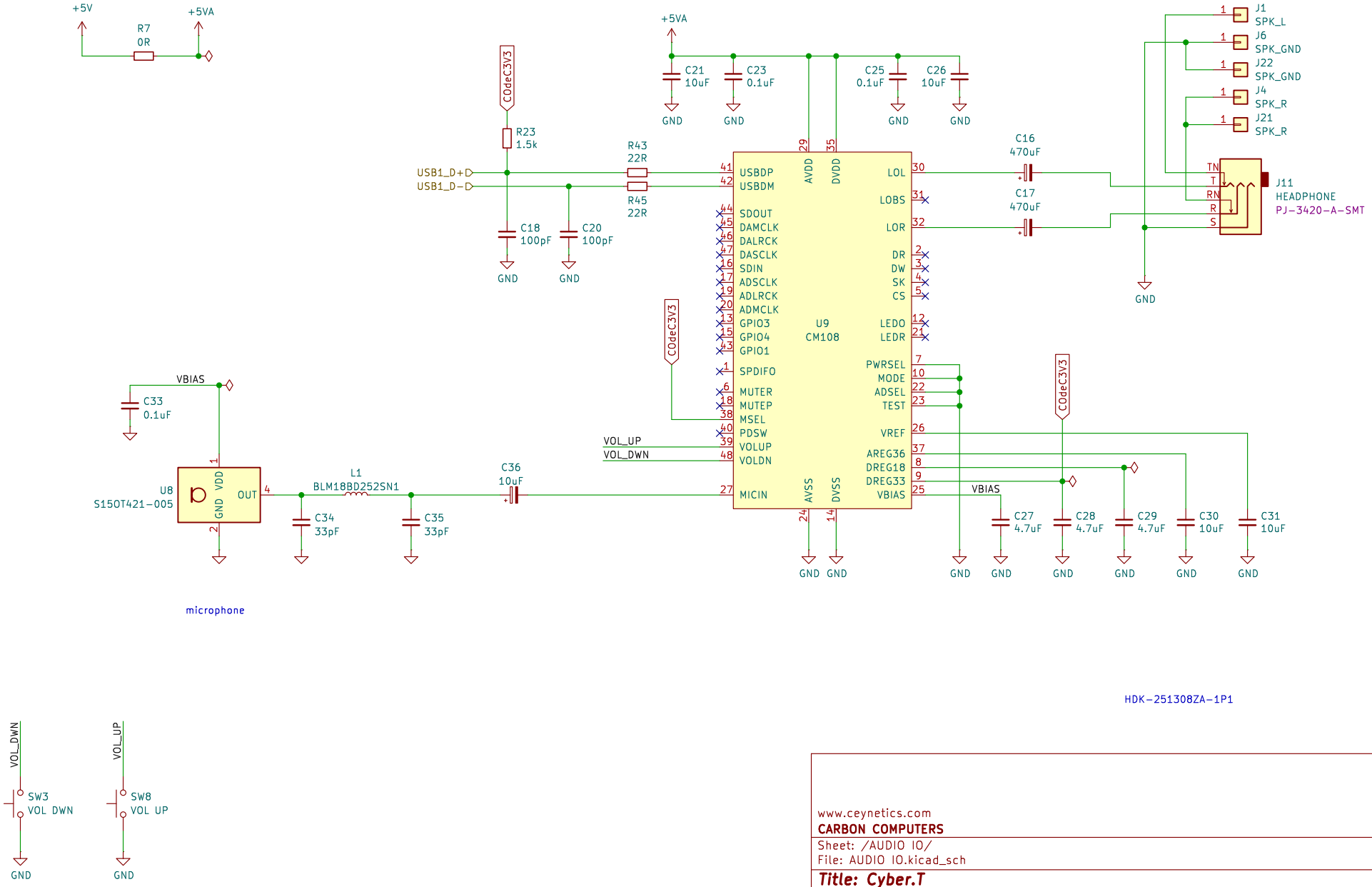
Supports Unified daughter board project

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Sheet: /USB HUB/
File: USB HUB.kicad_sch

Title: Cyber.T

Size: A3	Date: 2025-08-22	Rev: 3.0
KiCad E.D.A. 9.0.5		Id: 3/9



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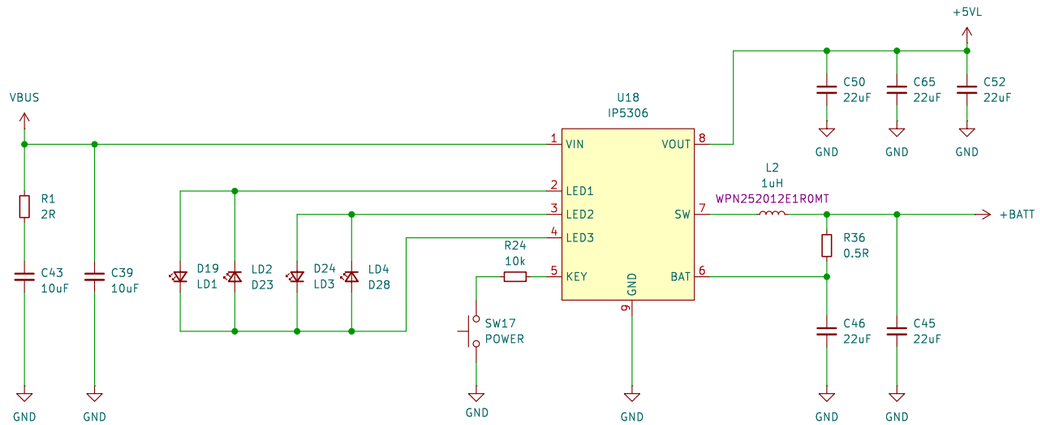
CARBON COMPUTERS

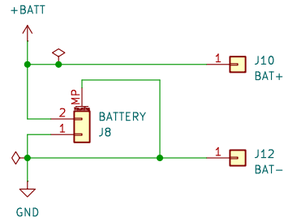
Sheet: /AUDIO IO/

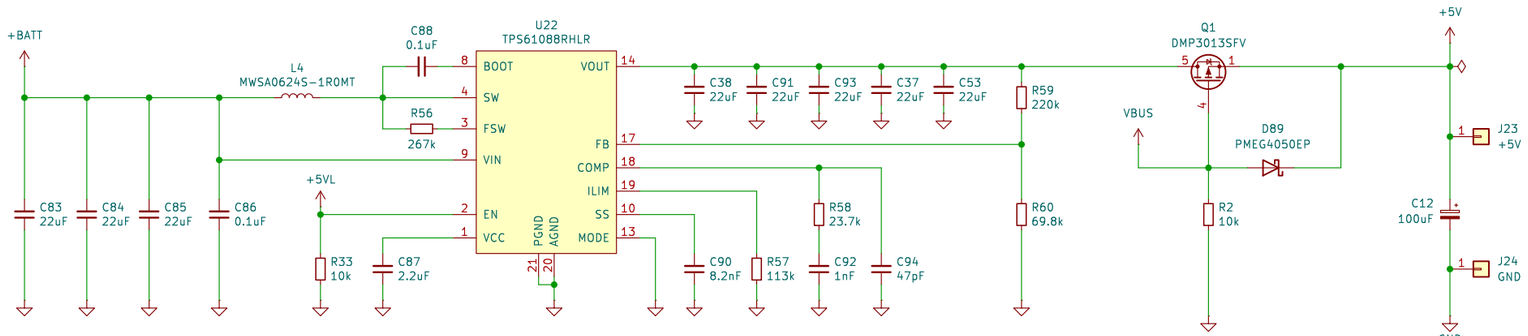
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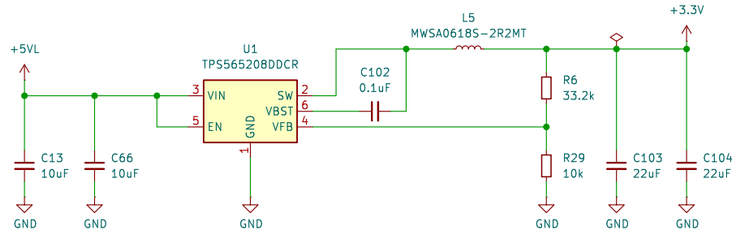
Title: Cyber.T

Size: A4	Date: 2025-08-22	**Rev: 3.0**
KiCad E.D.A. 9.0.5		Id: 4/9









General Purpose 3.3V Power Supply

3.3V @ 3A max

Symbol pinout & Symbol ERC
MPN & Footprint match/Schematic parity
Footprint parameters – for custom FP

Schematic connections

Power supply pins
Supply voltage
Power consumption

Reference schematic match
Component values
Component pinouts
Component footprints
Interface pinouts

Connection labels
Sheet to Sheet connections

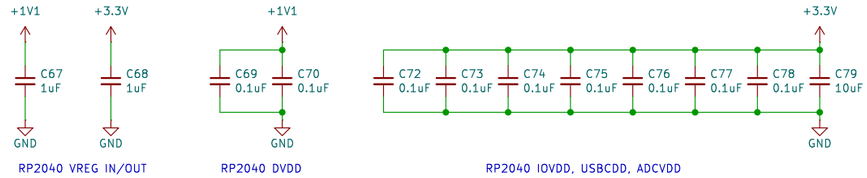
recheck all changes

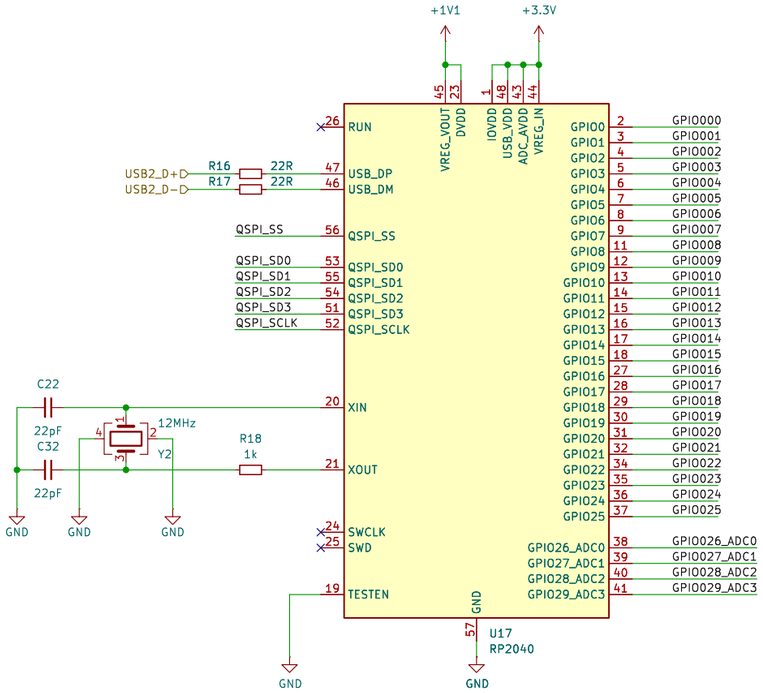
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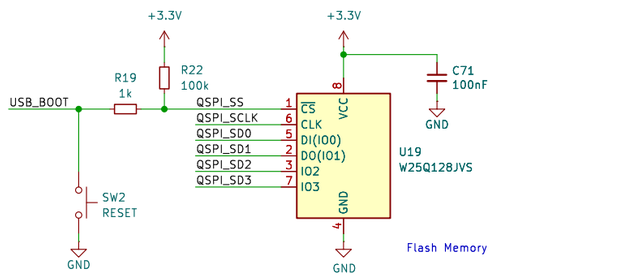
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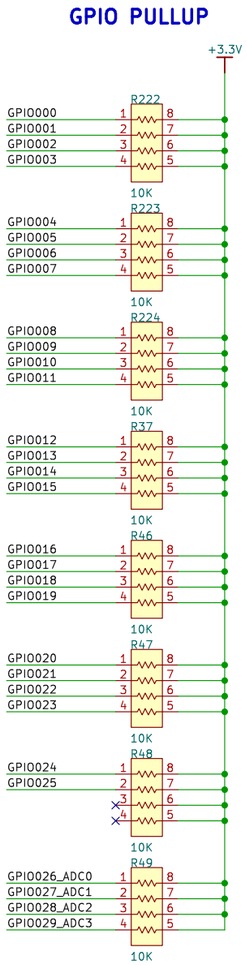
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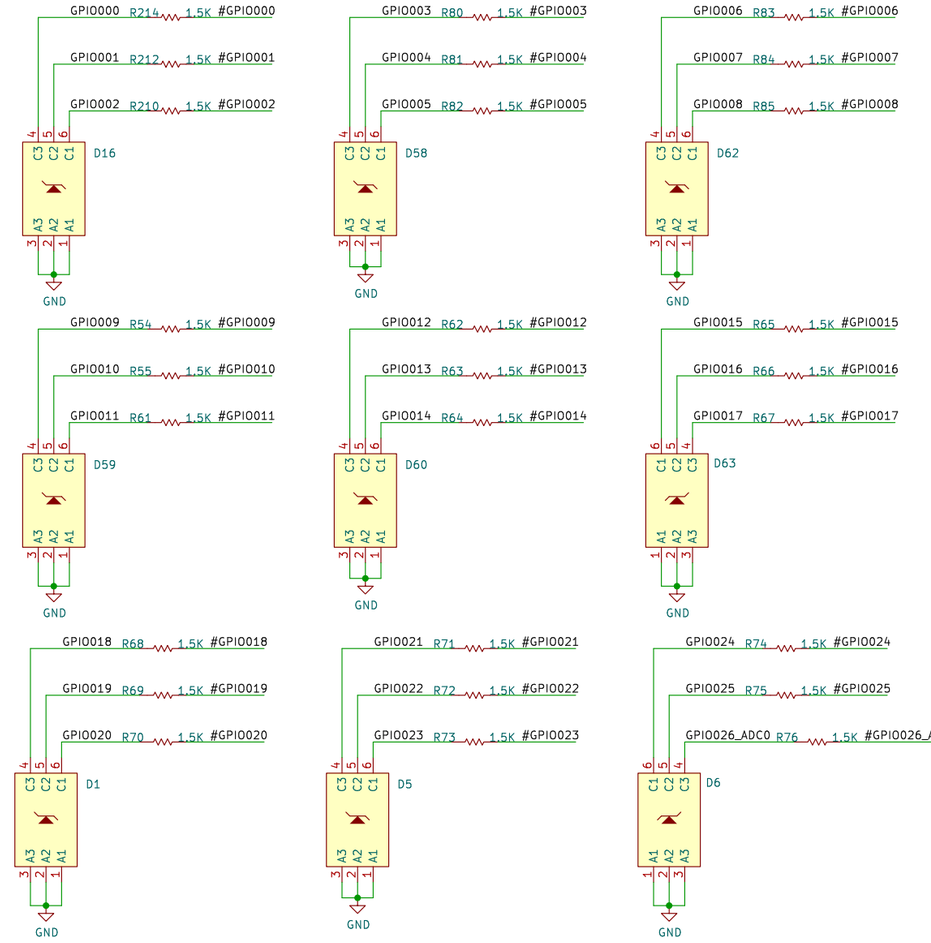
Size: A3	Date: 2025-08-22	**Rev: 3.0**
KiCad E.D.A. 9.0.5		Id: 5/9

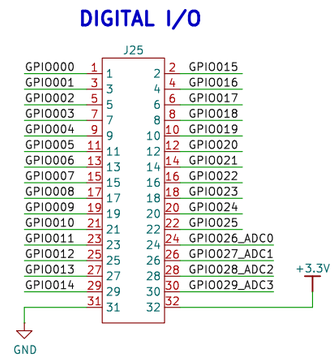












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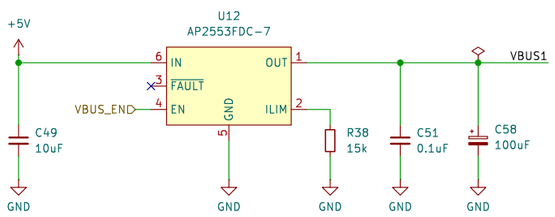
CARBON COMPUTERS

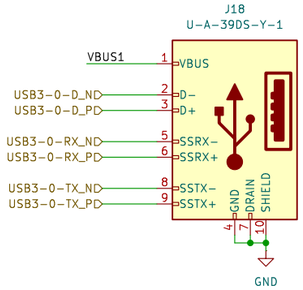
Sheet: /GPIO/

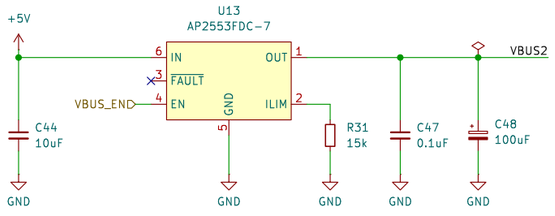
File: KEYPAD.kicad_sch

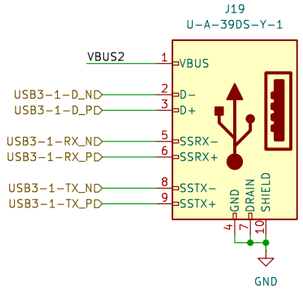
Title: Cyber.T

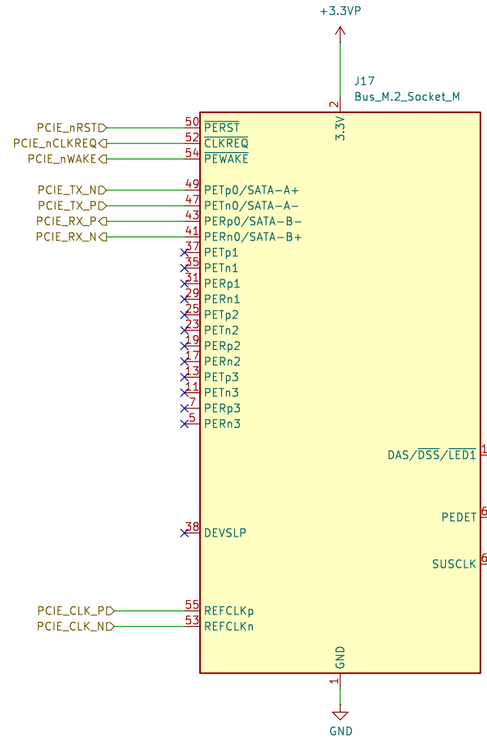
Size: A3	Date: 2025-08-22	**Rev: 3.0**
KiCad E.D.A. 9.0.5		Id: 6/9

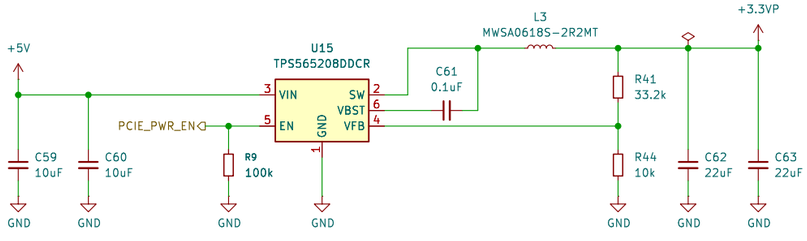








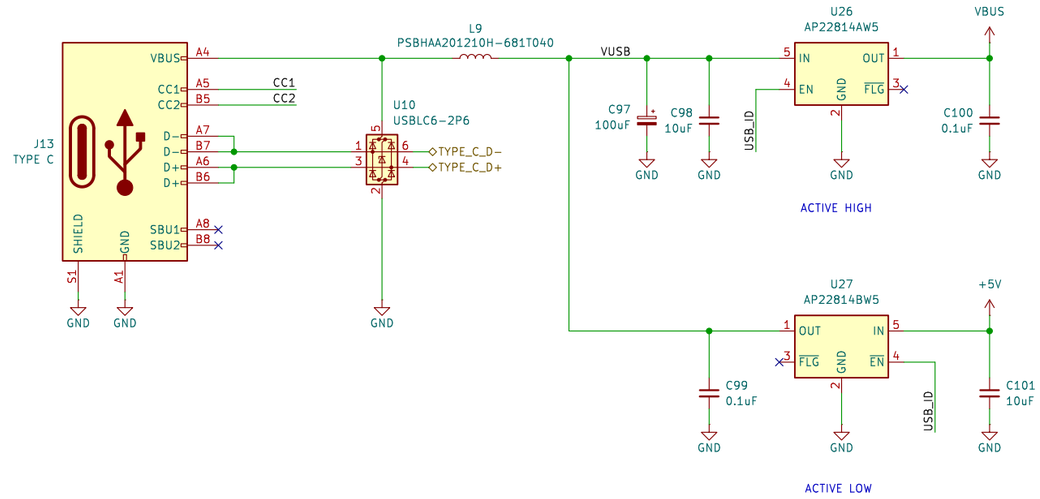


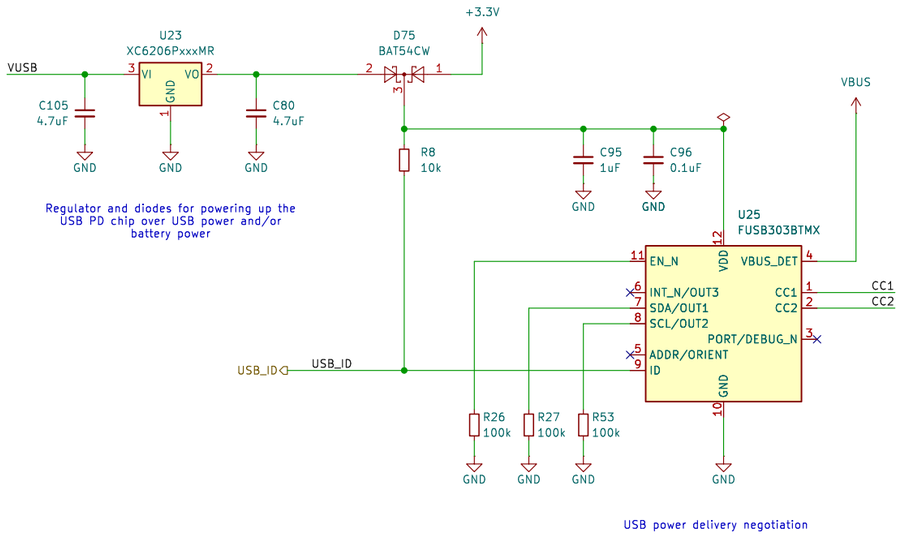


H5
KEY

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Sheet: /USB & PCIe/		
File: USB-PCIe.kicad_sch		
Title: Cyber.T		
Size: A3	Date: 2025-08-22	**Rev: 3.0**
KiCad E.D.A. 9.0.5		Id: 7/9





ID

LOW = FUSB303B attached as a Source
HIGH-Z = FUSB303B attached as a Sink

ADDR/ORIENT

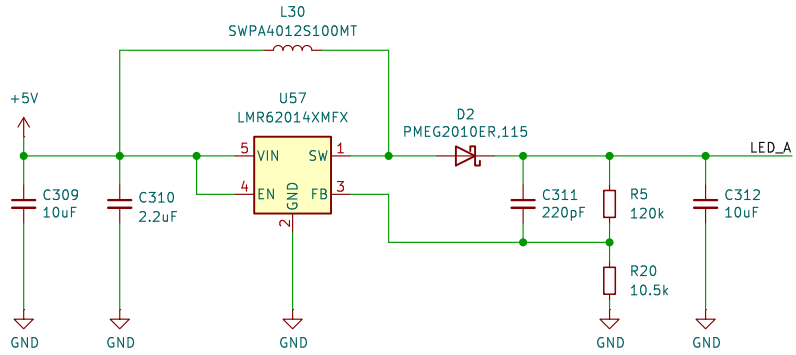
HIGH = I2 C mode with address 62h
Float = GPIO mode
LOW = I2 C mode with address 42h

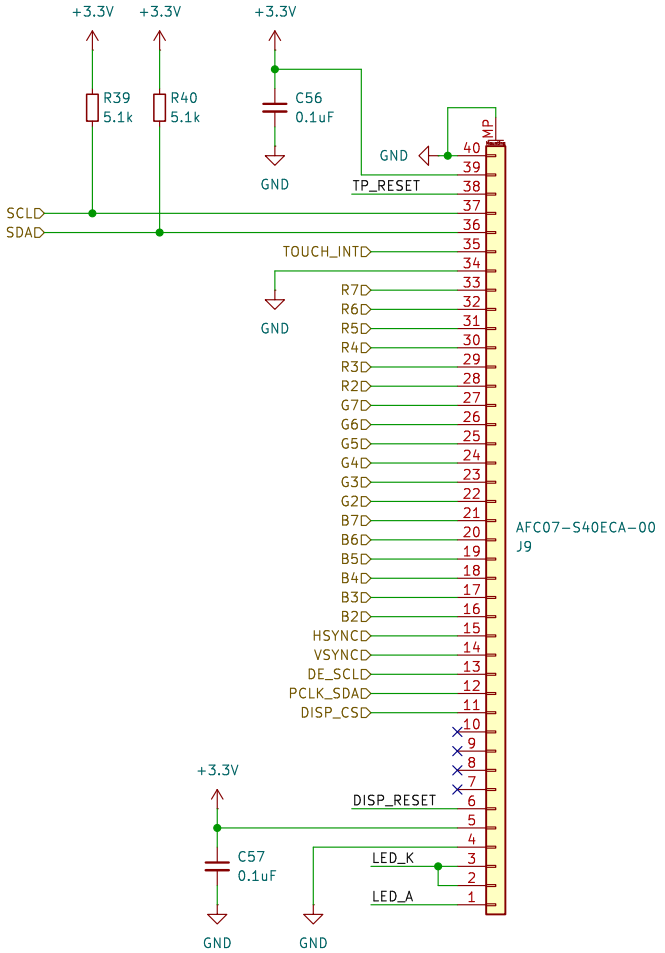
PORT/deBUG

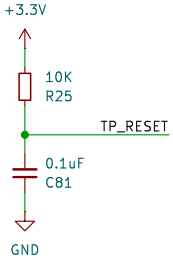
HIGH = FUSB303B as a Source Only port
Float = FUSB303B as a Dual Role Port (DRP)
LOW = FUSB303B as a Sink Only port

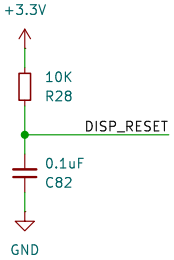
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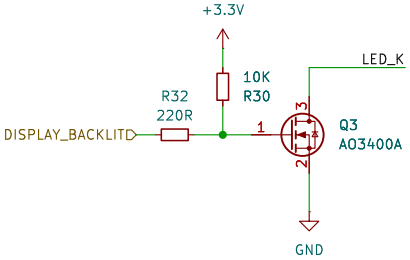
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File: USB CONNECTOR.kicad_sch		
Title: Cyber.T		
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KiCad E.D.A. 9.0.5		Id: 8/9











https://www.waveshare.com/wiki/2.8inch_DPI_LCD
https://pinout.xyz/pinout/hyperpixel4
https://github.com/raspberrypi/documentation/blob/develop/documentation/asciidoc/computers/raspberry-pi/display-parallel-interface.adoc

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Sheet: /DISPLAY/		
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Title: Cyber.T		
Size: A4	Date: 2025-08-22	**Rev: 3.0**
KiCad E.D.A. 9.0.5		Id: 9/9